

A source sends a bit $X \in \{0, 1\}$ with $\Pr(X = 0) = \frac{2}{5}$ over a binary channel with output $Y \in \{0, 1\}$, $W(1|0) = \frac{3}{4}$ and $W(0|1) = \frac{1}{2}$.

1. The receiver observes $y = 1$. What does the MAP detector decide?
 - (a) $\hat{x} = 0$
 - (b) $\hat{x} = 1$
 - (c) There is a tie between 0 and 1. *

Solution: By Definition 1.3 of the lecture notes, the MAP detector chooses the symbol with the largest joint probability $\Pr(x)W(y|x)$ for the observed output. For $y = 1$ the two joint probabilities are $\Pr(0)W(1|0) = \frac{2}{5} \cdot \frac{3}{4} = \frac{3}{10}$ and $\Pr(1)W(1|1) = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}$. They are equal, so there is a tie and the answer is (c). The likelihood favors 0 by a factor of $\frac{3}{2}$, the prior favors 1 by the same factor, and the two effects compensate exactly. Either decision gives the same average error probability.

2. The receiver uses the ML detector. What is its conditional error probability $P_e(0)$, when $x = 0$ is sent?
 - (a) $\frac{1}{4}$ *
 - (b) $\frac{3}{4}$
 - (c) $\frac{1}{10}$

Solution: By Definition 1.3 of the lecture notes, the ML detector chooses the symbol with the largest $W(y|x)$. For $y = 0$ we have $W(0|1) = \frac{1}{2} > W(0|0) = \frac{1}{4}$, so it decides $\hat{x} = 1$. For $y = 1$ we have $W(1|0) = \frac{3}{4} > W(1|1) = \frac{1}{2}$, so it decides $\hat{x} = 0$. The ML detector is not $\hat{x} = y$ on this channel, because it decides the opposite bit. When $x = 0$ is sent, the only wrong output is $y = 0$, and by equation (1.6) the conditional error probability is $P_e(0) = W(0|0) = \frac{1}{4}$. The answer is (a). Option (b) is $W(1|0)$, the error of the detector $\hat{x} = y$, which is not the ML detector here. Option (c) is the joint probability $\Pr(0)W(0|0)$, which includes the prior, whereas the conditional error probability of a given detector does not depend on the input distribution.

Seminar 1. Detectors and their errors

A symbol $X \in \{-1, 0, +1\}$ is sent over the real Gaussian channel with noise variance $\sigma^2 = \frac{1}{2}$. The probabilities are $\Pr(0) = \frac{1}{2}$ and $\Pr(-1) = \Pr(+1) = \frac{1}{4}$.

3. The receiver observes $y = \frac{1}{2}$. Draw the constellation and the channel output y on the real line. What does the MAP detector decide?
 - (a) $\hat{x} = 0$ *
 - (b) $\hat{x} = +1$
 - (c) There is a tie between 0 and +1.

Solution: By Definition 1.3 of the lecture notes, the MAP detector chooses the symbol with the largest $\Pr(x)W(y|x)$. On the real line, the constellation is the three points $-1, 0$ and $+1$, and the observation $y = \frac{1}{2}$ is the midpoint between 0 and $+1$. It is at the same distance $\frac{1}{2}$ from 0 and from $+1$, so the channel law of Table 1.2 gives the same value for both, $W(\frac{1}{2}|0) = W(\frac{1}{2}|+1)$. The prior of 0 is $\frac{1}{2}$ and the prior of $+1$ is $\frac{1}{4}$, so the product is twice as large for 0. The symbol -1 is farther, at distance $\frac{3}{2}$, so it is less likely, and its prior is also $\frac{1}{4}$. The MAP detector decides $\hat{x} = 0$, whatever the noise variance, and the answer is (a). Option (c) is the outcome of the nearest-point detector, which ignores the prior and has its threshold exactly at $y = \frac{1}{2}$.

4. The receiver uses the nearest-point detector. What is its conditional error probability $P_e(0)$, when $x = 0$ is sent?
 - (a) $\frac{1}{2} \operatorname{erfc}(\frac{1}{2})$
 - (b) $\operatorname{erfc}(1)$
 - (c) $\operatorname{erfc}(\frac{1}{2})$ *

Solution: The thresholds of the nearest-point detector of Proposition 1.5 are the midpoints $-\frac{1}{2}$ and $+\frac{1}{2}$, so the decision region of the symbol 0 is the interval $-\frac{1}{2} < y < \frac{1}{2}$. When 0 is sent the output is $Y = Z$, and an error occurs when $Z < -\frac{1}{2}$ or when $Z > \frac{1}{2}$. The two events are disjoint, so their probabilities add. By equation (1.23) of the lecture notes with the threshold $\frac{1}{2}$ and $\sqrt{2}\sigma = 1$, each of them is $\frac{1}{2} \operatorname{erfc}(\frac{1}{2})$. The sum of the two is $P_e(0) = \operatorname{erfc}(\frac{1}{2})$ and the answer is (c), as for the inner points of 4-PAM in Problem 1.12. Option (a) counts only one of the two sides. Option (b) uses the distance 1 to the neighbors instead of the distance $\frac{1}{2}$ to the thresholds. The prior does not appear, because the nearest-point detector does not use it and the error probability is conditional.

A QPSK symbol

$$X \in \left\{ \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2} \right\}$$

is sent over the complex Gaussian channel with noise variance σ^2 . The four symbols are equiprobable. The receiver uses the nearest-point detector.

5. The energy of a complex symbol x is $|x|^2 = \text{Re}(x)^2 + \text{Im}(x)^2$. What is the signal-to-noise ratio?

- (a) $\frac{1}{2\sigma^2}$
- (b) $\frac{1}{\sigma^2}$ *
- (c) $\frac{2}{\sigma^2}$

Solution: Following Section 1.2 of the lecture notes, the energy of each symbol is $\frac{1}{2} + \frac{1}{2} = 1$, because the real part and the imaginary part are both $\pm \frac{\sqrt{2}}{2}$. The average symbol energy is $E_s = 1$ and the signal-to-noise ratio is $\text{SNR} = E_s/\sigma^2 = 1/\sigma^2$. The answer is (b). Option (a) counts only one of the two components, and option (c) squares the sum $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}$ instead of adding the two squares.

6. The receiver observes an output y with $\text{Re}(y) > 0$ and $\text{Im}(y) < 0$. Draw the constellation and one such output y of your choice in the complex plane. What does the detector decide?

- (a) $\hat{x} = \frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}$ *
- (b) $\hat{x} = -\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}$
- (c) It depends on the values of $\text{Re}(y)$ and $\text{Im}(y)$.

Solution: The region of a symbol under the nearest-point detector is bounded by the perpendicular bisectors of the segments that join it to its neighbors, as explained in Section 1.5 of the lecture notes. For QPSK the bisectors are the real axis and the imaginary axis, so the four decision regions are the four quadrants, as in Figure 1.2(b) and Problem 1.13. The region of a symbol x is the quadrant where $\text{Re}(y)$ has the sign of $\text{Re}(x)$ and $\text{Im}(y)$ has the sign of $\text{Im}(x)$. In the complex plane, the constellation is the four corners of a square centered at the origin, one in each quadrant and all at distance 1 from the origin. An output with $\text{Re}(y) > 0$ and $\text{Im}(y) < 0$ is in the fourth quadrant, whose symbol is $\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}$, so the detector decides $\hat{x} = \frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}$. The answer is (a). Option (b) has the two signs reversed, and it is the farthest symbol from this output. Option (c) is not correct, because the two signs determine the quadrant, whatever the magnitudes of $\text{Re}(y)$ and $\text{Im}(y)$.

7. The symbol $x = \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}$ is sent. Its conditional error probability $P_e(x)$ is the probability that the channel output has

- (a) a negative real part and a negative imaginary part.
- (b) a negative real part or a negative imaginary part. *
- (c) a real part smaller than $\frac{\sqrt{2}}{2}$ or an imaginary part smaller than $\frac{\sqrt{2}}{2}$.

Solution: By equation (1.19) of the lecture notes, the conditional error probability is the probability that the output falls outside the decision region of the symbol sent. The region of $x = \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}$ is the first quadrant, $\text{Re}(y) > 0$ and $\text{Im}(y) > 0$. The output is outside it when at least one of the two conditions fails, that is, when $\text{Re}(y) < 0$ or $\text{Im}(y) < 0$. The answer is (b). Option (a) requires the two conditions to fail together, which is only the probability of deciding the opposite symbol $-\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}$. Option (c) places the thresholds at the symbol and not at the axes, and it gives $\frac{3}{4}$ for every SNR. The value follows from the correct decision, as in Problem 1.13. The two parts of the noise are independent, and each part of the output stays positive with probability $1 - \frac{1}{2} \text{erfc}(\frac{\sqrt{2}}{2\sigma})$, so $P_e(x) = 1 - \left(1 - \frac{1}{2} \text{erfc}(\frac{\sqrt{2}}{2\sigma})\right)^2$.